

Phenomenological Nature of Randomness in the TNA Framework

Statement

Within the TNA framework, randomness is not an ontological property of reality but a **phenomenological manifestation of structural constraints and observational limitations**.

Ontological level

At the ontological level, systems evolve within structural domains:

- state space Ω
- structural domain Ω_s
- structural boundary $\partial\Omega_s$

The system dynamics remain defined within these domains.

Therefore the evolution of the system is governed by **structural relations**, not by intrinsic randomness.

Phenomenological level

Randomness appears when:

- the observer has incomplete information about the system state
- the system is near structural bifurcations
- multiple admissible trajectories exist within the structural domain

In these situations, the system appears stochastic even if its evolution is structurally constrained.

Structural Interpretation

Randomness corresponds to **underdetermination at the observational level**, not to the absence of structure in reality.

Formally:

ontological level → structural dynamics
phenomenological level → probabilistic description

Probability therefore reflects **epistemic limitations or structural multiplicity**, not ontological indeterminacy.

Consequence

Within the TNA framework:

- the universe does not evolve through ontological randomness
- probabilistic descriptions arise from **limited access to structural information**

Thus randomness should be interpreted as **phenomenological**, not fundamental.

Short formulation

In the TNA framework, randomness is not a property of reality but a description of our limited access to structural dynamics.

Theorem: Structural Origin of Randomness

Definitions

Let

- Ω denote the global state space of a system.
- $\Omega_s \subseteq \Omega$ denote the **structural domain** in which the system dynamics are defined.
- $\partial\Omega_s$ denote the **structural boundary** of the domain.

Assume that the evolution of the system is governed by the dynamical equation

$$\dot{x} = F(x)$$

where

$$F : \Omega_s \rightarrow T(\Omega_s)$$

is a vector field defined on the structural domain.

Let $x(t)$ denote a trajectory satisfying

$$x(t) \in \Omega_s$$

for all admissible times t .

Theorem (Structural Origin of Randomness)

Apparent randomness emerges when trajectories approach the structural boundary $\partial\Omega_s$, where the internal dynamics cease to uniquely determine the continuation of the trajectory.

Proof

Inside the structural domain Ω_s , the system evolution is governed by the differential equation

$$\dot{x} = F(x)$$

with F defined on Ω_s .

Assume that F satisfies the regularity conditions required for existence and uniqueness of solutions of ordinary differential equations in the interior of the domain, for example local Lipschitz continuity.

Under these conditions, for any initial condition

$$x_0 \in \text{int}(\Omega_s)$$

there exists a unique trajectory

$$x(t)$$

satisfying the differential equation in a neighborhood of the initial state.

Thus, within the interior of the structural domain, the system evolution is uniquely determined by the dynamical law.

Now consider a trajectory approaching the structural boundary:

$$x(t) \rightarrow \partial\Omega_s$$

By definition, the boundary $\partial\Omega_s$ separates admissible states from states where the dynamical law F is not defined or not valid.

Therefore the vector field

$$F : \Omega_s \rightarrow T(\Omega_s)$$

does not extend beyond the structural boundary.

Consequently, the dynamical equation

$$\dot{x} = F(x)$$

cannot determine the continuation of the trajectory outside the domain.

Let x^* be a point on the boundary such that

$$x \setminus^* \in \partial\Omega_s$$

When the trajectory reaches an arbitrarily small neighborhood of $x \setminus^*$, the dynamical equation still determines the trajectory inside the domain but provides no rule for selecting a continuation beyond the boundary.

Since the state space Ω may contain multiple admissible structural domains adjacent to the boundary, multiple continuations of the trajectory may exist that satisfy admissibility constraints.

Thus the future evolution of the system cannot be uniquely determined by the internal dynamics defined on Ω_s .

An observer restricted to the dynamical description within Ω_s must therefore describe the possible continuations probabilistically.

This probabilistic description corresponds to the phenomenological appearance of randomness.

Therefore apparent randomness emerges from the structural indeterminacy associated with the boundary $\partial\Omega_s$, rather than from intrinsic stochasticity in the dynamics.

Corollary

Randomness observed in complex systems corresponds to **structural underdetermination near the boundaries of admissible dynamical domains**.

Conceptual Consequence

Within the TNA framework:

- deterministic evolution holds in the interior of structural domains
- indeterminacy arises when trajectories approach structural limits
- probabilistic descriptions emerge from the multiplicity of admissible continuations.

Thus randomness is **structural and phenomenological**, not ontological.

Theorem: Structural Origin of Randomness via Loss of Uniqueness

Definitions

Let

- Ω be the global state space of a dynamical system.
- $\Omega_s \subseteq \Omega$ be the **structural domain** where the system dynamics are defined.
- $\partial\Omega_s$ denote the boundary of the structural domain.

Assume the system evolution is governed by the differential equation

$$\dot{x} = F(x)$$

with

$$F : \Omega_s \rightarrow T(\Omega_s)$$

where $T(\Omega_s)$ denotes the tangent bundle of the domain.

Assume that F is locally Lipschitz continuous in the interior of Ω_s .

Theorem

If a trajectory approaches a structural boundary $\partial\Omega_s$ where the Lipschitz condition fails or the vector field cannot be uniquely extended, then the dynamical system may exhibit **loss of uniqueness of solutions**, resulting in multiple admissible future trajectories.

This multiplicity appears phenomenologically as randomness.

Proof

Consider an initial value problem

$$\dot{x} = F(x), \quad x(0) = x_0$$

with

$$x_0 \in \text{int}(\Omega_s)$$

Because F is locally Lipschitz continuous in the interior of the domain, the **Picard–Lindelöf theorem** ensures the existence and uniqueness of a solution in a neighborhood of the initial condition.

Thus for all states sufficiently far from the structural boundary, the system evolution is uniquely determined.

Now consider a trajectory $x(t)$ such that

$$x(t) \rightarrow x^*$$

with

$$x^* \in \partial\Omega_s$$

At the boundary, one of the following conditions may occur:

1. the vector field $F(x)$ ceases to be defined,
2. the Lipschitz condition fails,
3. the vector field admits multiple admissible extensions beyond the boundary.

In any of these cases, the assumptions required for the Picard–Lindelöf uniqueness theorem no longer hold.

Therefore uniqueness of solutions cannot be guaranteed.

Consequently, for states arbitrarily close to x^* , there may exist multiple trajectories

$$x_1(t), x_2(t), \dots$$

that satisfy the differential equation and share the same limiting state.

This loss of uniqueness corresponds mathematically to a **bifurcation of trajectories** in the state space.

From the perspective of an observer who only knows the system dynamics inside the structural domain Ω_s , there is no deterministic rule to select among the admissible continuations.

Therefore the evolution must be described probabilistically.

Thus apparent randomness arises from the multiplicity of admissible solutions near the structural boundary.

Corollary

Randomness in deterministic systems can arise from **structural bifurcations associated with loss of uniqueness of solutions near domain boundaries**.

Interpretation

Within the TNA framework:

- the interior of the structural domain admits deterministic evolution
 - the structural boundary introduces potential bifurcations
 - multiplicity of admissible trajectories produces phenomenological randomness.
-

Conceptual Summary

Formally:

```
interior of  $\Omega_s$  → Lipschitz dynamics → unique trajectory  
near  $\partial\Omega_s$  → loss of Lipschitz condition → multiple trajectories
```

Observed randomness corresponds to the inability to determine which admissible trajectory the system will follow.

Theorem: Structural Boundary Implies Dynamic Bifurcation Possibility

Definitions

Let

- Ω be the global state space of a dynamical system.
- $\Omega_s \subseteq \Omega$ be a structural domain in which the system dynamics are defined.
- $\partial\Omega_s$ denote the structural boundary.

Assume the system evolution is governed by

$$\dot{x} = F(x)$$

with

$$F : \Omega_s \rightarrow T(\Omega_s)$$

and F locally Lipschitz continuous in the interior of Ω_s .

Theorem

If a structural boundary $\partial\Omega_s$ exists, then there exist states arbitrarily close to that boundary for which the dynamical system admits multiple admissible continuations.

Therefore the existence of a structural boundary implies the possibility of **dynamic bifurcation**.

Proof

Inside the interior of the structural domain

$$x \in \text{int}(\Omega_s)$$

the system evolution satisfies

$$\dot{x} = F(x)$$

with F locally Lipschitz continuous.

By the **existence and uniqueness theorem for ordinary differential equations**, the trajectory through any interior point is unique.

Now consider a point

$$x^* \in \partial\Omega_s$$

Since x^* lies on the boundary of the structural domain, at least one of the following must hold:

1. the vector field $F(x)$ is not defined beyond the boundary,
2. the dynamical law changes across the boundary,
3. admissible system states belong to different structural regimes.

Therefore the dynamical law governing the system cannot be uniquely extended across $\partial\Omega_s$ while preserving the same structural description.

Now consider a sequence of points

$$x_n \in \Omega_s$$

such that

$$x_n \rightarrow x^*$$

Because the dynamical law loses structural validity at the boundary, arbitrarily small perturbations of the state near x^* may correspond to trajectories entering different admissible regimes of the state space.

Thus there exist states arbitrarily close to the boundary whose trajectories diverge toward different regions of Ω .

This divergence corresponds to **bifurcation of trajectories**.

Therefore the presence of a structural boundary implies the possibility of dynamic bifurcation.

Corollary

Structural limits naturally generate regions where system trajectories become sensitive to perturbations and may branch into multiple admissible dynamical evolutions.

Interpretation

In the TNA framework:

```
interior of  $\Omega_S \rightarrow$  deterministic evolution  
approach to  $\partial\Omega_S \rightarrow$  structural indeterminacy  
structural indeterminacy  $\rightarrow$  bifurcation of trajectories
```

Thus structural limits are natural sources of:

- bifurcations
 - sensitivity to perturbations
 - phenomenological randomness.
-

Conceptual consequence

This result connects three pieces of the theory:

1. **Structural Non-Derivability**
2. **Structural Origin of Randomness**
3. **Dynamic Bifurcation at Structural Limits**

Theorem: Structural Collapse Theorem

Definitions

Let

- Ω be the global state space of a system.
- $\Omega_s \subseteq \Omega$ be a structural domain where the system dynamics are defined.
- $\partial\Omega_s$ be the structural boundary of that domain.

Assume the system evolves according to

$$\dot{x} = F_s(x)$$

with

$$F_s : \Omega_s \rightarrow T(\Omega_s)$$

representing the dynamical law valid within the structural domain.

Theorem (Structural Collapse)

If a trajectory reaches the structural boundary $\partial\Omega_s$, the system cannot continue evolving under the same structural dynamics and must transition to a different structural regime.

Proof

Assume the system state evolves within the structural domain

$$x(t) \in \Omega_s$$

according to the dynamical law

$$\dot{x} = F_s(x)$$

defined on Ω_s .

Because F_s is defined only on Ω_s , the evolution of the system is valid only while the state remains inside this domain.

Now consider a time t^* such that

$$x(t^*) \in \partial\Omega_s$$

that is, the trajectory reaches the structural boundary.

By definition of structural boundary:

- states beyond $\partial\Omega_s$ do not satisfy the structural conditions that define the domain, or
- the dynamical law F_s is no longer valid beyond the boundary.

Therefore the differential equation

$$\dot{x} = F_s(x)$$

cannot determine the system evolution outside Ω_s .

If the system state were to continue evolving, it must satisfy a different dynamical law

$$\dot{x} = F_{s'}(x)$$

defined on another structural domain

$$\Omega_{s'} \subseteq \Omega.$$

Thus continuation of the trajectory necessarily implies transition from

$$\Omega_s \rightarrow \Omega_{s'}$$

which corresponds to a **change of structural regime**.

If no such domain exists that admits continuation of the trajectory, the system loses structural coherence and the trajectory terminates within the admissible state space.

Therefore reaching the structural boundary implies structural collapse of the original dynamical regime.



Corollary

Structural collapse corresponds to the **transition between structural domains in the global state space**.

Interpretation

Within the TNA framework:

trajectory inside $\Omega_S \rightarrow$ stable structural regime
trajectory reaches $\partial\Omega_S \rightarrow$ structural collapse
post-collapse $\rightarrow$ transition to $\Omega_{S'}$

Structural Phase Diagram of the Dynamic Universe

Concept

Within the TNA framework, the global state space of reality can be represented as a collection of **structural domains** embedded in a larger dynamical landscape.

Each domain corresponds to a regime where a specific structural organization and dynamical law remain valid.

Formally:

- Ω : global state space
 - $\Omega_s^1, \Omega_s^2, \dots$: structural domains
 - $\partial\Omega_s$: structural boundaries separating regimes
-

Structural Regions

The structural phase diagram contains three fundamental regions.

1. Stable Structural Domains

Inside a domain Ω_s :

- the governing dynamics are well-defined
- trajectories are stable
- the system exhibits deterministic evolution.

These correspond to **stable regimes of organization**.

Examples:

- stable ecosystems
- coherent biological organisms

- functioning economies
 - stable physical phases.
-

2. Boundary Regions

Near structural boundaries $\partial\Omega_s$:

- the system becomes sensitive to perturbations
- trajectory multiplicity emerges
- phenomenological randomness appears.

These regions correspond to **structural instability zones**.

3. Collapse / Transition Zones

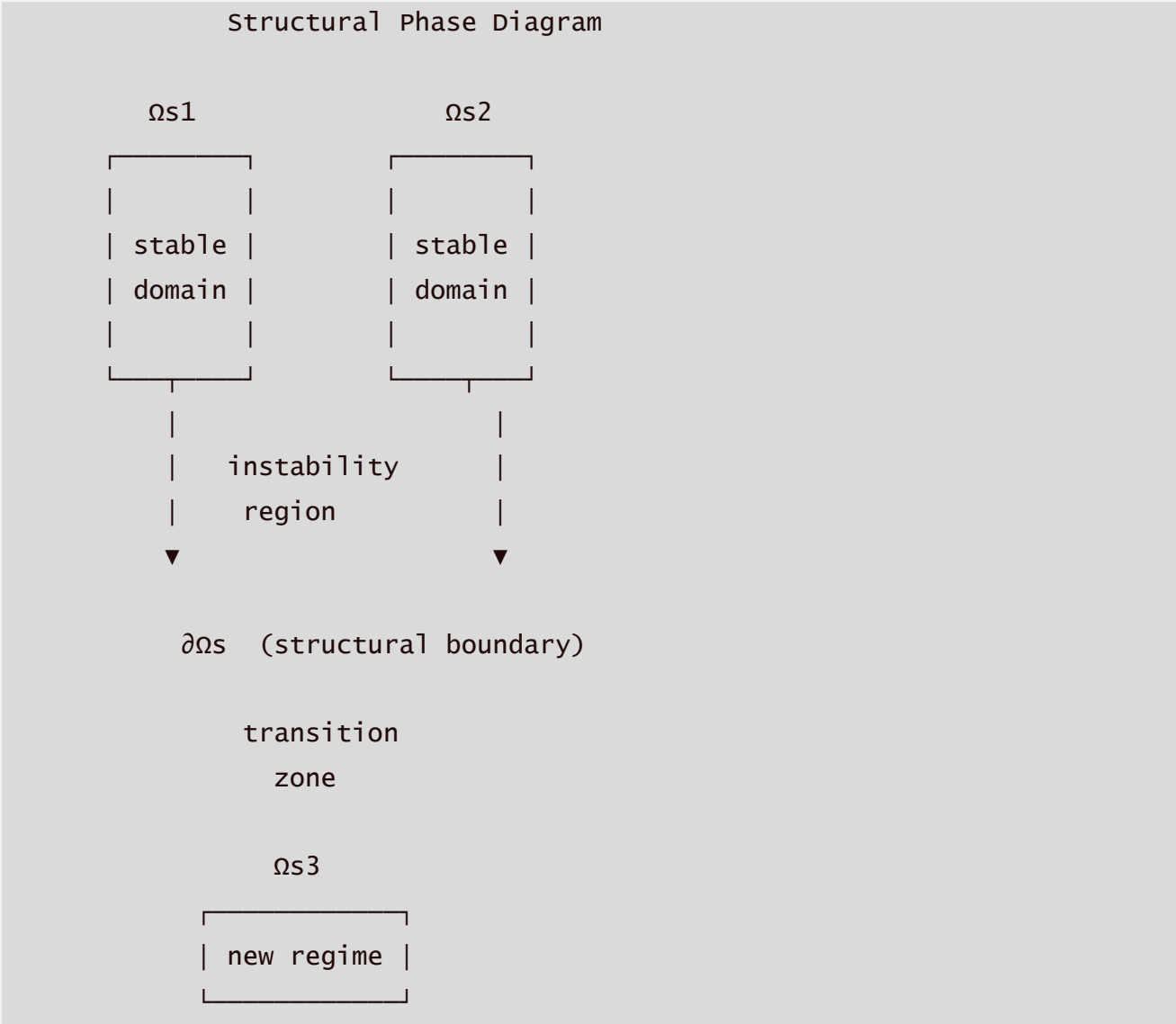
When trajectories reach structural boundaries:

- the original structural regime collapses
- the system must transition to another structural domain.

This corresponds to **structural phase transitions**.

Global Diagram

Conceptually, the universe can be represented as a landscape of structural domains separated by boundaries.



Structural Interpretation

The evolution of complex systems can therefore be described as motion through this structural landscape.

Three phases appear naturally:

Connection with Previous Theorems

The structural phase diagram integrates the main results of the TNA framework.

Structural Boundary Implies Dynamic Bifurcation

→ explains trajectory branching near boundaries.

Structural Origin of Randomness

→ explains probabilistic behavior in instability zones.

Structural Collapse Theorem

→ explains transitions between domains.

Key Consequence

The structural phase diagram implies that many phenomena across disciplines follow the same pattern:

1. stable structural regime
 2. increasing instability and fluctuations
 3. structural collapse or transition
 4. emergence of a new regime.
-

Interpretation

This suggests that reality is not best understood as a single dynamical system, but as a **network of structural domains connected through transitions**.

In this view, evolution across physics, biology, and social systems corresponds to navigation across the structural phase diagram.

Structural Phase Diagram of the Dynamic Universe

